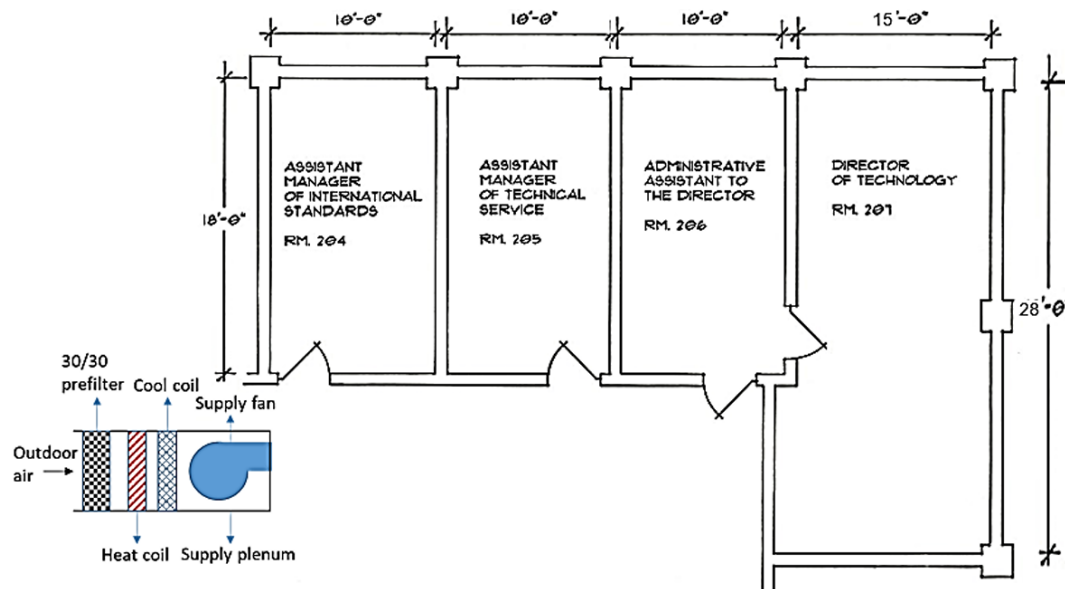


Question 1 (2.3)

A second floor of an office building has been given below:



An HVAC engineer has determined the amount of air per room required to maintain a comfortable and quiet space

Office Space	Heating Air Requirement (cfm)
Office 204	330
Office 205	470
Office 206	190
Office 207	520

a) The architect has requested the design of a round ductwork system (duct material, sizes of round duct, necessary fittings, including diffusers, in a ductwork system, and the 2D system layout). You can use the dimensions of rooms to estimate the duct lengths. Note: You are free to design a duct system layout (No specific guidance needs to follow).

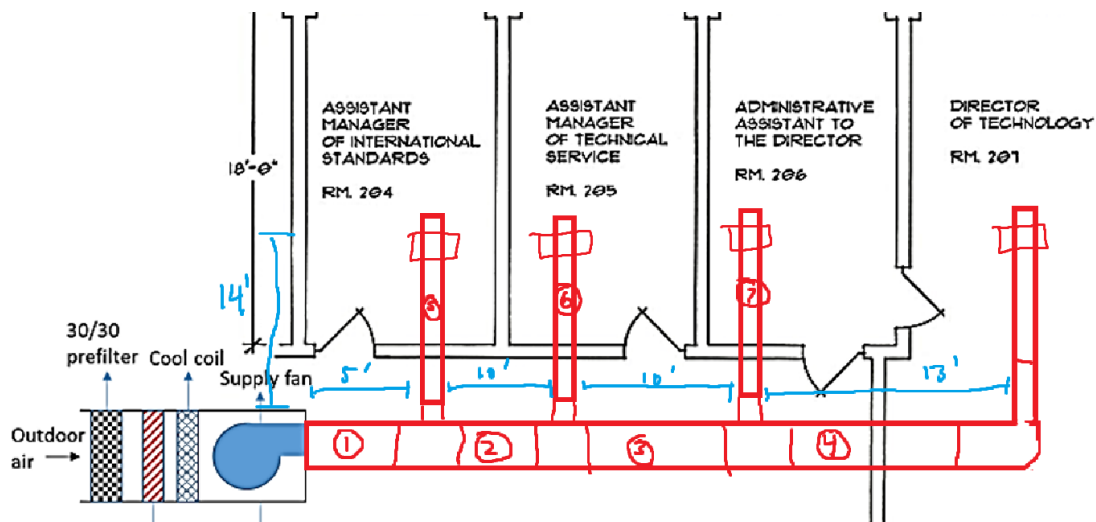
1 Assumptions:

1. $v_{\max} = 1200$ fpm since the air is to be quiet for the office space.
2. Round ductwork system

3. ΔP unknown $\rightarrow$ friction loss = 0.1" wg/100 ft.
4. Diffusers are to be placed in each room. Assume that the pressure loss across the diffuser is 0.05" wg. (Table 2.2).
5. Galvanized steel duct material is used (common in air ducts).
6. Bellmouth entrance from supply fan and diffuser entrance are to be considered.
7. The elbow is pleated 90°.

2 Sketches

Let us draw a sketch of a possible ductwork system layout for the office building.



3 Analysis

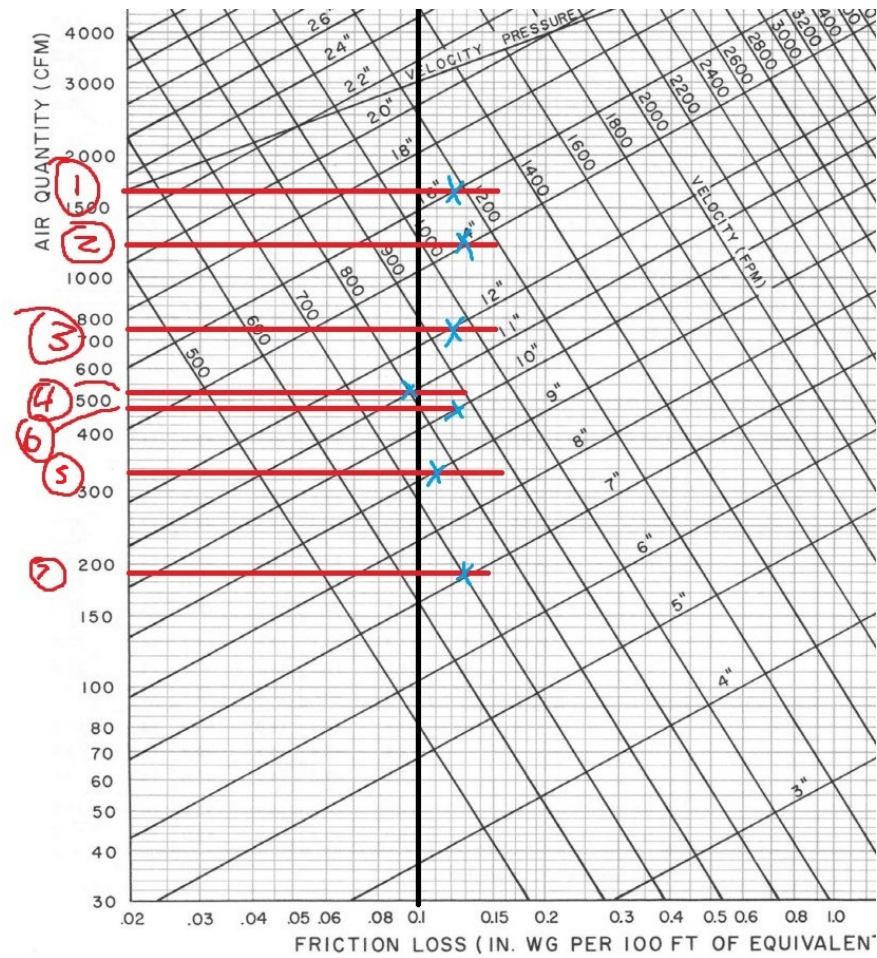
3.1 CFM in Sections

Let us determine the required cfm for the differens sections. Based on the required cfm from the rooms, one can set:

Segment	Required cfm, V_i
5	330
6	470
7	190
4	520
3	$V_7 + V_4 = 710$
2	$V_6 + V_3 = 1180$
1	$V_5 + V_2 = 1510$

3.2 Section Diameter

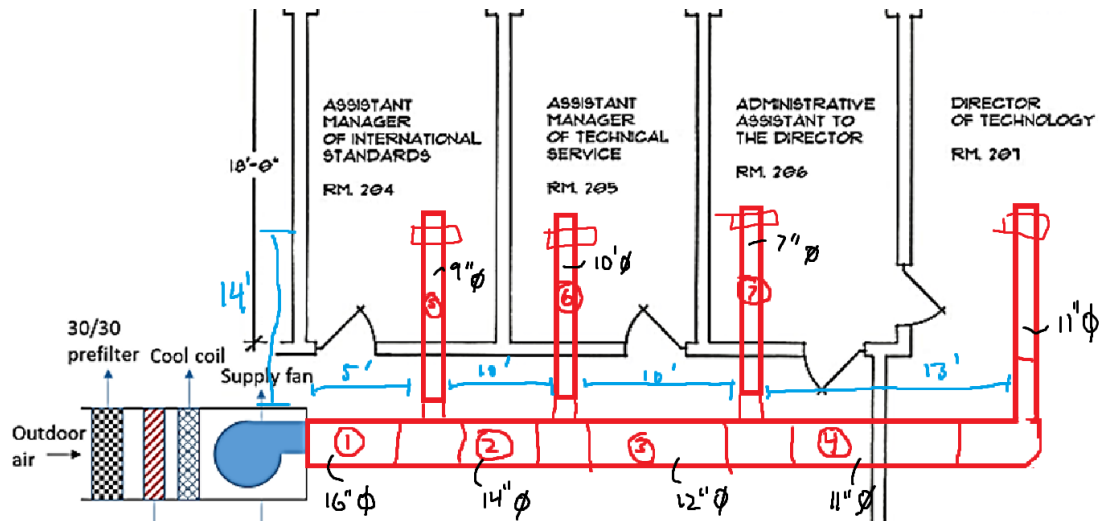
Using the required cfm and the friction loss, the diameter of the ducts can be determined from the sizing chart:



In summary,

Segment	Required cfm	Diameter	Air Velocity fpm	Friction Loss (in. wg/100ft)
1	1510	16"	1150	0.12
2	1180	14"	1100	0.13
3	710	12"	950	0.12
4	520	11"	800	0.095
5	330	9"	750	0.11
6	470	10"	850	0.12
7	190	7"	700	0.13

4 Final Drawing



5 Conclusion

The system has been designed with round ducts with velocities that do not exceed the maximum velocity of 1200 fpm. The friction losses were within an acceptable range of 0.1" wg/100 ft. The ducts were sized according to the required cfm and the pressure drop was calculated for each segment. The final drawing shows the layout of the ductwork system for the office building. Dampers can be added at the diffusers to balance the airflow in each room.

b) Based on the ductwork design, calculate the fan's minimum operating condition (airflow rate and static pressure). To be closer to reality, pressure losses from the air handling unit, including a prefilter, heating coils, cooling coils, and supply plenum, need to be considered in this project. The air handling unit is located in the bottom left corner, as indicated in this sketch.

1 Assumptions

Assume Table 2.2 and A.4 are valid and applicable to the project.

2 Analysis

2.1 Equivalent Lengths

First determine the bellmouth equivalent lengths from Table A.4:

$$L_{\text{BM},1} = (16'') \cdot (12) = 192'' = 16'$$

$$L_{\text{BM},5} = (9'') \cdot (12) = 120'' = 9'$$

$$L_{\text{BM},6} = (10'') \cdot (12) = 120'' = 10'$$

$$L_{\text{BM},7} = (7'') \cdot (12) = 84'' = 7'$$

$$L_{\text{BM},4} = (11'') \cdot (12) = 132'' = 11'$$

Next, determine the tee equivalent lengths from Table A.4:

$$L_{(\text{tee, thru},1)} = (16'') \cdot (8) = 128'' = 10.67'$$

$$L_{(\text{tee, thru},2)} = (14'') \cdot (8) = 112'' = 9.33'$$

$$L_{(\text{tee, thru},3)} = (12'') \cdot (8) = 96'' = 8'$$

$$L_{(\text{tee, thru},4)} = (11'') \cdot (8) = 88'' = 7.33'$$

$$L_{(\text{tee, branch},5)} = (9'') \cdot (8) = 72'' = 6'$$

$$L_{(\text{tee, branch},6)} = (10'') \cdot (8) = 80'' = 6.67'$$

$$L_{(\text{tee, branch},7)} = (7'') \cdot (8) = 56'' = 4.67'$$

And lastly the elbow equivalent length from Table A.4:

$$L_{\text{elbow, } 90^\circ, 4} = (11'') \cdot (15) = 165'' = 13.75'$$

The total equivalent for relevant sections are:

$$\begin{aligned}
 L_{1 \rightarrow 2} &= L_{\text{BM},1} + L_1 + L_{(\text{tee}, \text{thru},1)} \\
 &= 16' + 5' + 10.67' = 31.67' \\
 L_{2 \rightarrow 3} &= L_2 + L_{(\text{tee}, \text{thru},2)} \\
 &= 14' + 9.33' = 23.33' \\
 L_{3 \rightarrow 4} &= L_3 + L_{(\text{tee}, \text{thru},3)} \\
 &= 10' + 8' = 18' \\
 L_{2 \rightarrow 204} &= L_{\text{tee}, \text{branch},5} + L_5 + L_{\text{BM},5} \\
 &= 6' + 14' + 9' = 29' \\
 L_{3 \rightarrow 205} &= L_{\text{tee}, \text{branch},6} + L_6 + L_{\text{BM},6} \\
 &= 6.67' + 14' + 10' = 30.67' \\
 L_{4 \rightarrow 206} &= L_{\text{tee}, \text{branch},7} + L_7 + L_{\text{BM},7} \\
 &= 4.67' + 14' + 7' = 25.67' \\
 L_{4 \rightarrow 207} &= L_4 + L_{\text{elbow}, 90^\circ, 4} + L_{4,h} + L_{\text{BM},4} \\
 &= 11' + 13.75' + 5' + 11' = 40.75'
 \end{aligned}$$

2.2 Pressure Drop

Converting to pressure drop:

$$\begin{aligned}
 \Delta P_{1 \rightarrow 2} &= \frac{0.12}{100} \cdot 31.67 = 0.038 \text{ in-wg} \\
 \Delta P_{2 \rightarrow 3} &= \frac{0.13}{100} \cdot 23.33 = 0.030 \text{ in-wg} \\
 \Delta P_{3 \rightarrow 4} &= \frac{0.12}{100} \cdot 18 = 0.022 \text{ in-wg} \\
 \Delta P_{2 \rightarrow 204} &= \frac{0.11}{100} \cdot 29 = 0.032 \text{ in-wg} \\
 \Delta P_{3 \rightarrow 205} &= \frac{0.12}{100} \cdot 30.67 = 0.037 \text{ in-wg} \\
 \Delta P_{4 \rightarrow 206} &= \frac{0.13}{100} \cdot 25.67 = 0.033 \text{ in-wg} \\
 \Delta P_{4 \rightarrow 207} &= \frac{0.095}{100} \cdot 40.75 = 0.039 \text{ in-wg}
 \end{aligned}$$

The pressure drop across the air handling unit can be calculated using Table 2.2:

$$\begin{aligned}
 \Delta P_{\text{handling}} &= \Delta P_{\text{plenum}} + \Delta P_{\text{filter}} + \Delta P_{\text{heating}} + \Delta P_{\text{cooling}} \\
 &= 0.50 + 0.30 + 0.35 + 0.35 = 1.50 \text{ in-wg}
 \end{aligned}$$

The total pressure drop is:

$$\begin{aligned}\Delta P_{\text{handling} \rightarrow 204} &= \Delta P_{1 \rightarrow 2} + \Delta P_{2 \rightarrow 204} + \Delta P_{\text{handling}} + \Delta P_{\text{diffuser}} \\ &= 0.038 + 0.032 + 1.50 + 0.05 = 1.62 \text{ in-wg}\end{aligned}$$

$$\begin{aligned}\Delta P_{\text{handling} \rightarrow 205} &= \Delta P_{1 \rightarrow 2} + \Delta P_{2 \rightarrow 3} + \Delta P_{3 \rightarrow 205} + \Delta P_{\text{handling}} + \Delta P_{\text{diffuser}} \\ &= 0.038 + 0.030 + 0.037 + 1.50 + 0.05 = 1.65 \text{ in-wg}\end{aligned}$$

$$\begin{aligned}\Delta P_{\text{handling} \rightarrow 206} &= \Delta P_{1 \rightarrow 2} + \Delta P_{2 \rightarrow 3} + \Delta P_{3 \rightarrow 4} + \Delta P_{4 \rightarrow 206} + \Delta P_{\text{handling}} + \Delta P_{\text{diffuser}} \\ &= 0.038 + 0.030 + 0.022 + 0.033 + 1.50 + 0.05 = 1.67 \text{ in-wg}\end{aligned}$$

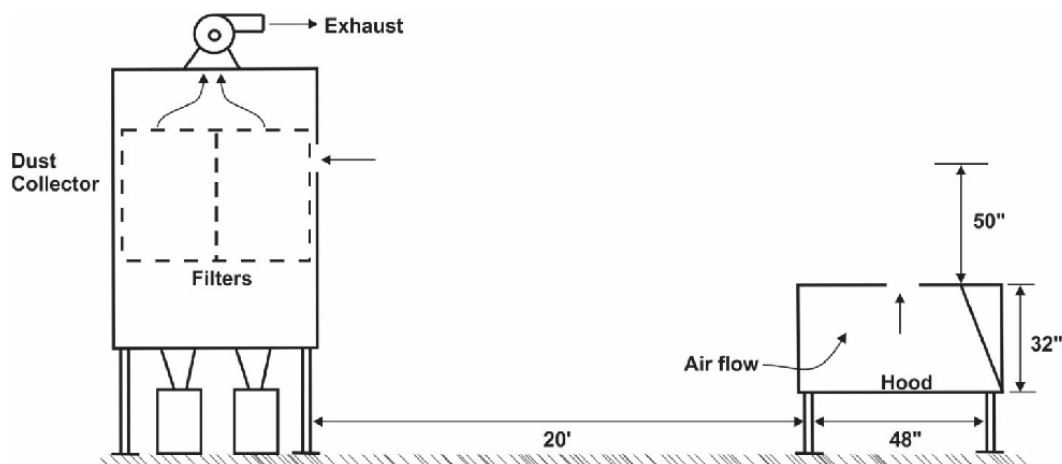
$$\begin{aligned}\Delta P_{\text{handling} \rightarrow 207} &= \Delta P_{1 \rightarrow 2} + \Delta P_{2 \rightarrow 3} + \Delta P_{3 \rightarrow 4} + \Delta P_{4 \rightarrow 207} + \Delta P_{\text{handling}} + \Delta P_{\text{diffuser}} \\ &= 0.038 + 0.030 + 0.022 + 0.039 + 1.50 + 0.05 = 1.68 \text{ in-wg}\end{aligned}$$

3 Conclusion

The greatest pressure drop is through the air handling unit to Office 207. Therefore minimum operating condition for the fan is 1.68" in. wg and 1510 cfm.

Question 2 (2.9)

The National Research Council has decided to pursue research in the area of spray-dried agglomeration of nano-sized powder particles to produce micron-sized powder particles. Safety and health regulations permit only a limited amount of these particles to escape into the ambient air of the space. To that end, the Council has contracted the services of Alliance Engineering Corp. to design a high-velocity duct system using a Farr® Gold Series 10 dust collector. The dust collector will draw 4000 cfm of air to eliminate powder particles from the space. The air will be drawn through a hood and the dust collector, as shown below, with dimensions. The dust collector and hood have a total pressure loss of 2.7 in. w.g. The Council will provide a lab environment with a maximum duct velocity of 2500 to 6000 fpm.

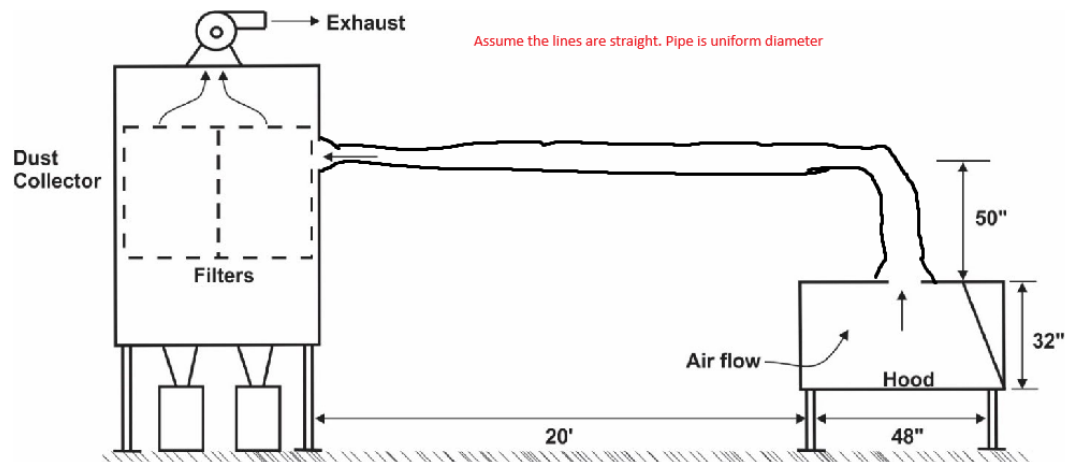


a) Design and size round ductwork between the hood and the dust collector (the duct layout, accessories, and fittings).

1 Assumptions

1. Bellmouth used at both the hood and the dust collector.
2. Round ductwork system
3. The elbow is pleated 90°.
4. Galvanized steel duct material is used (common in air ducts).
5. The airflow is 4000 cfm.
6. The pressure loss for the hood and dust collector is 2.7 in. w.g.

2 Sketches



3 Analysis

3.1 Determine Air Velocity

From Table 2.3

Table 2.3 Maximum supply duct velocities

High-Velocity Systems (Main Branches)	
Airflow Rate in Duct (cfm)	Maximum Velocity (fpm)
1000–3000	2500
3000–6000	3000
6000–10000	3500
10000–15000	4000
15000–25000	4500
25000–40000	5000
40000–60000	6000

Source: Howell, Sauer, and Coad [9].

It is clear the maximum velocity is 3000 fpm. This is within the range of 2500 to 6000 fpm.

3.2 Determine Duct Diameter

Using A.1:

Appendix A

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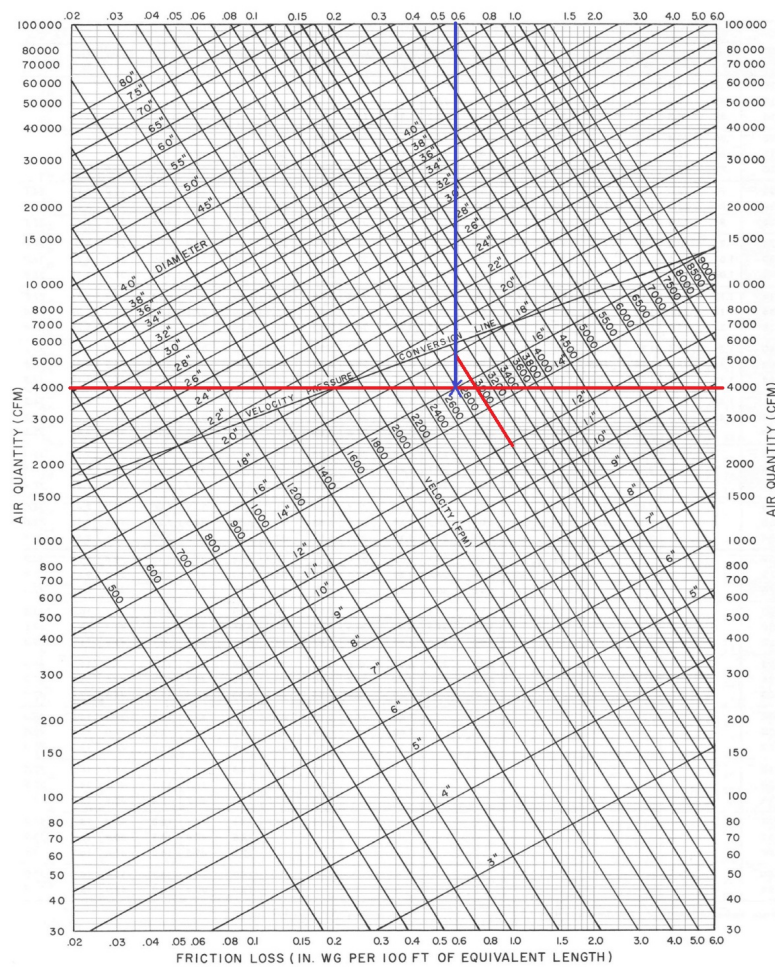


Figure A.1 Friction Loss in Round (Straight) Ducts. Source: *System Design Manual, Part 2: Air Distribution*, Carrier Air Conditioning Co., Syracuse, NY, 1974 (Reprinted with permission)

From the chart, 16", 2800 fpm, 0.6 in. wg. per 100ft. is the closest to the design conditions.

3.3 Determine Equivalent Length

The length of the straight portions are

$$L_{\text{straight}} = 20' + \frac{48''}{2} + 50'' = 20' + 2' + 4.17' = 26.17'$$

The equivalent length of the elbow is

$$L_{\text{elbow}} = 16'' \cdot 15 = 240'' = 20'$$

The equivalent length of the bellmouths are

$$L_{\text{bellmouth, hood}} = L_{\text{bellmouth, col}} = 16'' \cdot 12 = 192'' = 16'$$

The total equivalent length is

$$\begin{aligned} L_{\text{total}} &= L_{\text{straight}} + L_{\text{elbow}} + L_{\text{bellmouth, hood}} + L_{\text{bellmouth, col}} \\ &= 26.17' + 20' + 16' + 16' = 78.17' \end{aligned}$$

3.4 Determine Pressure Loss

The pressure drop from the duct is

$$\Delta P_{\text{duct}} = \frac{0.6}{100} \cdot 78.17' = 0.47 \text{ in. w.g.}$$

The total pressure drop is

$$\begin{aligned} \Delta P_{\text{total}} &= \Delta P_{\text{duct}} + \Delta P_{\text{hood+col}} \\ &= 0.47 + 2.7 \\ &= 3.17 \text{ in. w.g.} \end{aligned}$$

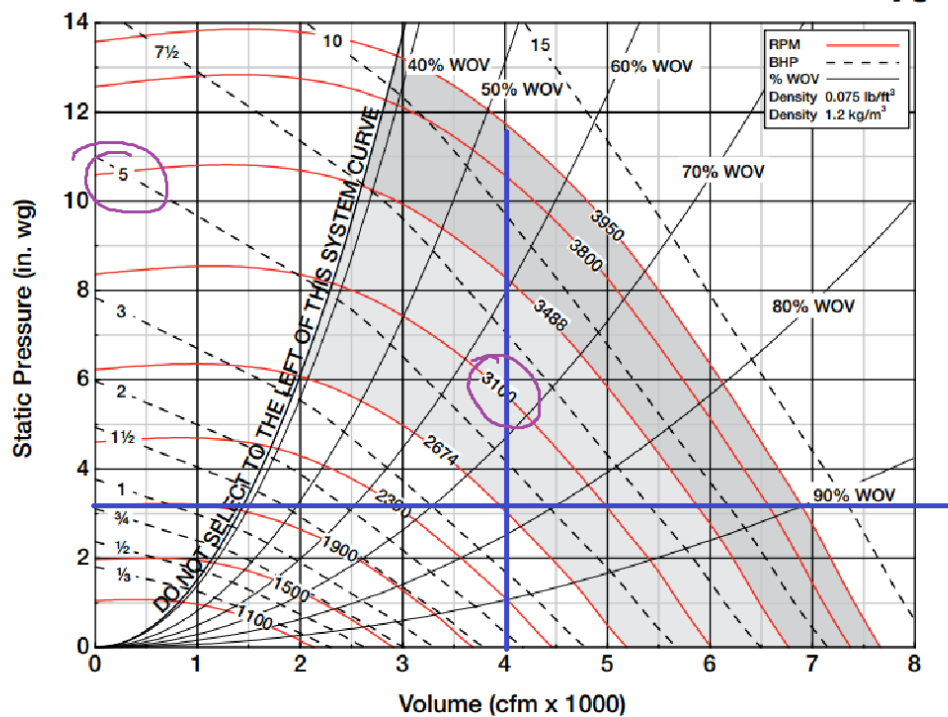
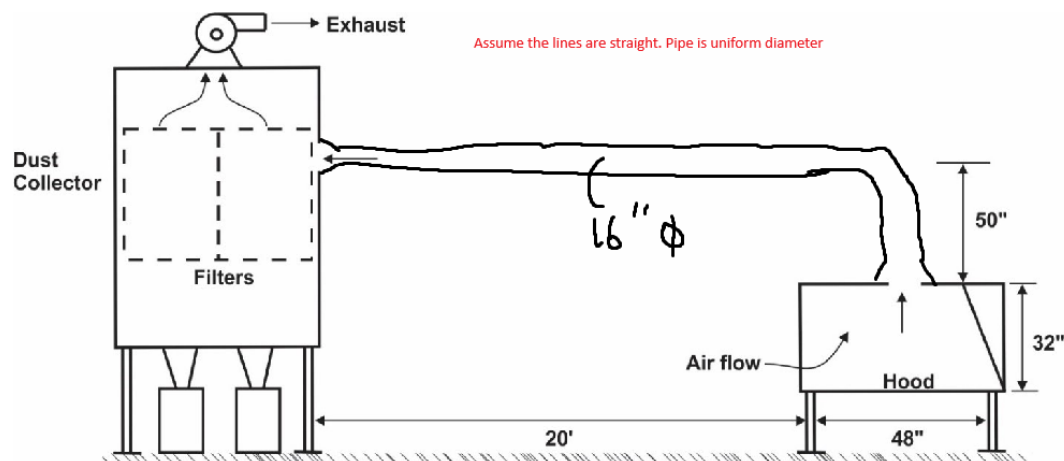
4 Conclusion

The ductwork should be 16" in diameter. The fan should be able to handle a pressure drop of 3.17 in. w.g. and a flow rate of 4000 cfm.

b) *Select an appropriate Greenheck 15 BISW fan (identify nominal fan speed and motor power).*

1 Fan Selection

The fan should move at least 4000 cfm and support a pressure drop of 3.17 in. w.g. Based on the performance curves, a nominal fan of 3100 rpm and 5 hp motor is the smallest specification that meets the requirements.

15 BISW**2 Drawings****3 Conclusion**

A round duct of 16" has been selected for this project based on a required airflow of 4000 cfm and a maximum air velocity of 3000 fpm.

A pleated 90° elbow is used to connect the hood to the dust collector. A bellmouth is used at both the hood and the dust collector. A smooth 90° elbow would prevent dust from accumulating in the ductwork and should be considered in further work.